

Yousef A. Alsalloum

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EDUCATION

University of Southern California

Master of Science, Cybersecurity Engineering

Los Angeles, CA

May 2028

The Pennsylvania State University

Bachelor of Science, Cybersecurity Analytics and Operations

Magna Cum Laude

University Park, PA

May 2026

GPA: 3.93/4.00

SKILLS

Security and Analysis: Kali Linux, Wireshark, Ghidra Decompiler, HxD Hex Editor, Decryption Techniques, Ciphey, StegPy, StegCracker, OSINT

Programming Languages: Python, JavaScript, C#, Java, HTML, CSS

Tools and Platforms: Ubuntu Linux, Git, GitHub, Microsoft Windows, MacOS

Other: UI/UX Design, Frontend Development, Fluent in Arabic and English

EXPERIENCE

Undergraduate Research Assistant

The Pennsylvania State University

January 2025 – March 2026

University Park, PA

- Built 2 custom Python tools to collect and archive social media data, including a CLI that scraped hundreds of thousands of tweets and a Wayback Machine tool that archived hundreds of web pages.
- Automated retrieval of posts, user metadata, and page snapshots to support analysis of diplomatic social media usage.
- Curated datasets and analyzed engagement and messaging patterns among state-affiliated accounts.

Learning Assistant

The Pennsylvania State University

January 2025 – May 2025

University Park, PA

- Supported instruction for Discrete Mathematics (IST 230) through weekly office hours and in-class concept clarification.
- Collaborated with 2 peers to lead concept reviews and strengthen student understanding across key topics.
- Graded assignments and supported the professor with assessment planning and course logistics.

PROJECTS

X-Search-Scraper

Twitter/X Data Collection Tool

- Developed a Python CLI integrating an X (formerly Twitter) data scraping service to automate retrieval of tweets and user metadata via public endpoints and archival methods for OSINT and research applications.
- Designed with JSON/CSV export support and modular architecture for adaptation to changing APIs.

Wayback-SS-Tool

Web Archive Screenshot Automation

- Built a Python CLI that automates screenshot capture of archived web pages from the Wayback Machine using a Selenium-powered headless browser, archiving hundreds of pages across specified URLs and timestamps.
- Implemented configurable output options including image width, filename formatting, directory control, and overwrite handling for research and forensic backup use cases.

Wazza

Chat Application with End-to-End Encryption

- Developed a secure real-time chat application using Python (FastAPI) and JavaScript with full client-side end-to-end encryption via Elliptic-Curve Diffie-Hellman key exchange and AES-256-GCM.
- Implemented a browser-native encryption workflow where keys never leave the client and all WebSocket communication transmits ciphertext only.
- Designed with modular encryption logic, automatic key negotiation, and extensibility toward Signal-style forward secrecy and multi-user rooms.